

Visualization

```
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.4.3

Warning: package 'ggplot2' was built under R version 4.4.3

Warning: package 'lubridate' was built under R version 4.4.2

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2    4.0.1    ✓ tibble     3.2.1
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.0.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
data <- read_csv("train_update.csv")
```

Rows: 14113 Columns: 11

— Column specification —

Delimiter: ","

chr (3): Pipe ID, Material, Surface

dbl (5): GPS x1, GPS y1, GPS x2, GPS y2, Cost

date (2): Lay date, Date

time (1): Time

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
set.seed(900)
na.omit(data)
```

A tibble: 14,107 × 11

	`Pipe ID`	`Lay date`	Material	Surface	`GPS x1`	`GPS y1`	`GPS x2`	`GPS y2`
	<chr>	<date>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	P19711	1962-09-17	wrought iron	farmla...	1822.	3189.	1790.	3202.
2	P07628	1965-03-30	gray iron	grassl...	4062.	2607.	4007	2589.
3	P31195	1948-04-13	wrought iron	grassl...	3009.	2473.	3006.	2459.
4	P15715	1965-12-16	wrought iron	struct...	3051.	3478.	3052.	3483.
5	P10415	1953-06-13	wrought iron	swamp	3573.	4657.	3591.	4648.
6	P19170	1948-05-18	wrought iron	water	2918.	1941.	2919.	1955.
7	P03320	1955-02-19	copper	struct...	4179.	6636.	4157.	6636.

```

8 P27701 1964-08-29 wrought iron water 1563. 2392. 1570. 2403.
9 P42748 1953-04-17 wrought iron swamp 3321. 4809. 3313. 4814.
10 P12943 1954-03-08 gray iron swamp 3834. 3134. 3836. 3143.
# i 14,097 more rows
# i 3 more variables: Date <date>, Time <time>, Cost <dbl>

```

```

data<-cbind(data,as.numeric((data$Date-data$`Lay date`)/365))

colnames(data)[12] <- "Years"

data<-cbind(data,sqrt((data$`GPS y2`-data$`GPS y1`)^2 +
                      (data$`GPS x2`-data$`GPS x1`)^2))
colnames(data)[13] <- "Length"
head(data)

```

```

Pipe ID Lay date Material Surface GPS x1 GPS y1 GPS x2 GPS y2
1 P19711 1962-09-17 wrought iron farmland 1821.64 3189.18 1789.59 3201.56
2 P07628 1965-03-30 gray iron grassland 4062.27 2606.80 4007.00 2588.85
3 P31195 1948-04-13 wrought iron grassland 3009.11 2473.32 3005.81 2458.59
4 P15715 1965-12-16 wrought iron structure 3050.60 3478.42 3051.62 3482.92
5 P10415 1953-06-13 wrought iron swamp 3573.19 4657.26 3591.42 4648.12
6 P19170 1948-05-18 wrought iron water 2918.40 1940.81 2919.23 1954.63

Date Time Cost Years Length
1 2019-01-06 08:20:00 2074.90 56.34247 34.357923
2 2019-01-07 00:00:00 618.71 53.81096 58.111749
3 2019-04-16 03:48:00 1050.07 71.05479 15.095128
4 2019-04-16 16:51:00 1000.81 53.36712 4.614152
5 2019-04-17 22:08:00 3157.83 65.88767 20.392952
6 2019-06-19 17:07:00 312.59 71.13425 13.844902

```

```

sf1 <- 0.7
sf2 <- 0.15
library(ggplot2)
# Logic:
# 1. randomly permute (reorder) numbers 1,2,...,n
# 2. take positions of first 70% as training set "set=1"
# 3. take positions of next 15% as validation set "set=2"
# 4. take positions of last 15% as test set "set=3"
reorder <- sample.int(n = nrow(data), size = nrow(data), replace = FALSE)

set <- ifelse(test = (reorder < sf1 * nrow(data)), yes = 1,
             no = ifelse(test = (sf1 * nrow(data) < reorder & reorder < (sf1 + sf2)
                           * nrow(data)), yes = 2, no = 3
             )
)
val <- data[set == 2,]
fit1 <- lm(Years~Surface+Material+Length+`GPS x1`+`GPS x2`+`GPS y1`+`GPS y2`, data=data[set==1,])
fit2 <- lm(Years~Surface+Material+Length, data=data[set==1,])

pred_v_s1 <- predict(fit1, newdata = val)

```

```
MSPE_v_s1 <- mean((as.numeric(val$Years) - pred_v_s1)^2)
pred_v_s2 <- predict(fit2, newdata = val)
MSPE_v_s2 <- mean((as.numeric(val[,12]) - pred_v_s2)^2)
summary(fit1)
```

Call:

```
lm(formula = Years ~ Surface + Material + Length + `GPS x1` +
  `GPS x2` + `GPS y1` + `GPS y2`, data = data[set == 1, ])
```

Residuals:

Min	1Q	Median	3Q	Max
-30.1209	-1.3604	-0.2507	1.1545	18.9236

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	64.6454734	0.3518345	183.738	< 2e-16 ***
Surfacegrassland	-0.0407397	0.1671220	-0.244	0.807412
Surfaceroad	0.0002242	0.2426533	0.001	0.999263
Surfacestructure	-1.2003436	0.2169625	-5.532	3.24e-08 ***
Surfaceswamp	0.1700720	0.2511776	0.677	0.498359
Surfacewater	-0.5446712	0.1631153	-3.339	0.000843 ***
Materialcast iron	-1.2256353	0.2476161	-4.950	7.55e-07 ***
Materialcopper	4.1154673	0.3144213	13.089	< 2e-16 ***
Materialgray iron	2.0594918	0.2898711	7.105	1.29e-12 ***
Materialwrought iron	0.2944986	0.2286127	1.288	0.197707
Length	-0.0825016	0.0043529	-18.953	< 2e-16 ***
`GPS x1`	-0.0065780	0.0029075	-2.262	0.023694 *
`GPS x2`	0.0070288	0.0029074	2.418	0.015644 *
`GPS y1`	0.0014480	0.0024934	0.581	0.561446
`GPS y2`	-0.0017908	0.0024933	-0.718	0.472628

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.865 on 9858 degrees of freedom

(6 observations deleted due to missingness)

Multiple R-squared: 0.1399, Adjusted R-squared: 0.1387

F-statistic: 114.5 on 14 and 9858 DF, p-value: < 2.2e-16

```
summary(fit2)
```

Call:

```
lm(formula = Years ~ Surface + Material + Length, data = data[set ==
  1, ])
```

Residuals:

Min	1Q	Median	3Q	Max
-29.8780	-1.3980	-0.2551	1.0815	20.0683

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	65.043221	0.256926	253.159	< 2e-16	***
Surfacegrassland	0.425359	0.163545	2.601	0.009313	**
Surfaceroad	0.410267	0.243146	1.687	0.091572	.
Surfacestructure	-0.752028	0.211605	-3.554	0.000381	***
Surfaceswamp	0.372268	0.243601	1.528	0.126498	
Surfacewater	-0.210495	0.160574	-1.311	0.189923	
Materialcast iron	-1.254502	0.246441	-5.090	3.64e-07	***
Materialcopper	3.808718	0.303158	12.563	< 2e-16	***
Materialgray iron	2.339726	0.292829	7.990	1.50e-15	***
Materialwrought iron	-0.250883	0.224320	-1.118	0.263416	
Length	-0.087652	0.003968	-22.091	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.926 on 9862 degrees of freedom

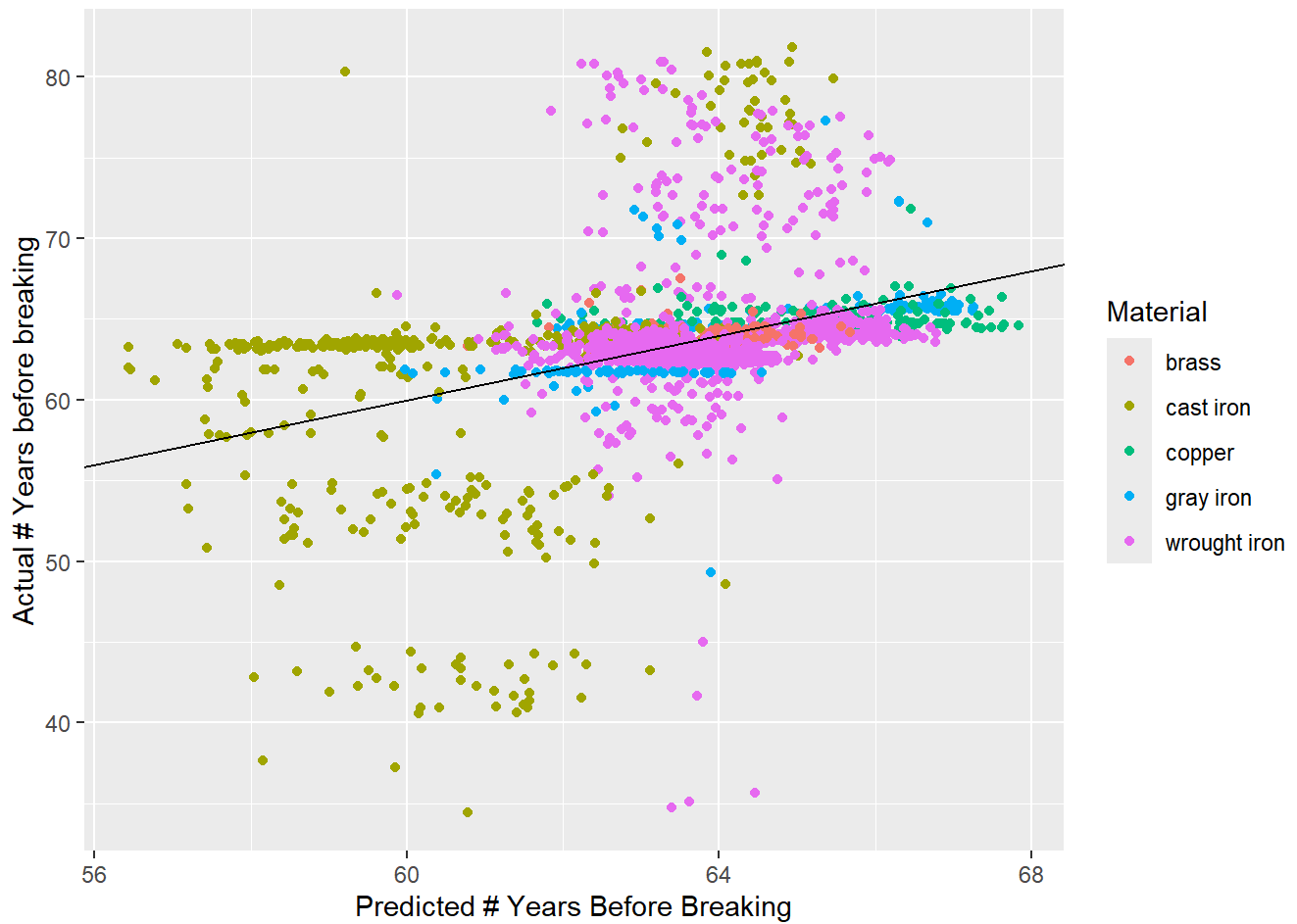
(6 observations deleted due to missingness)

Multiple R-squared: 0.1179, Adjusted R-squared: 0.117

F-statistic: 131.8 on 10 and 9862 DF, p-value: < 2.2e-16

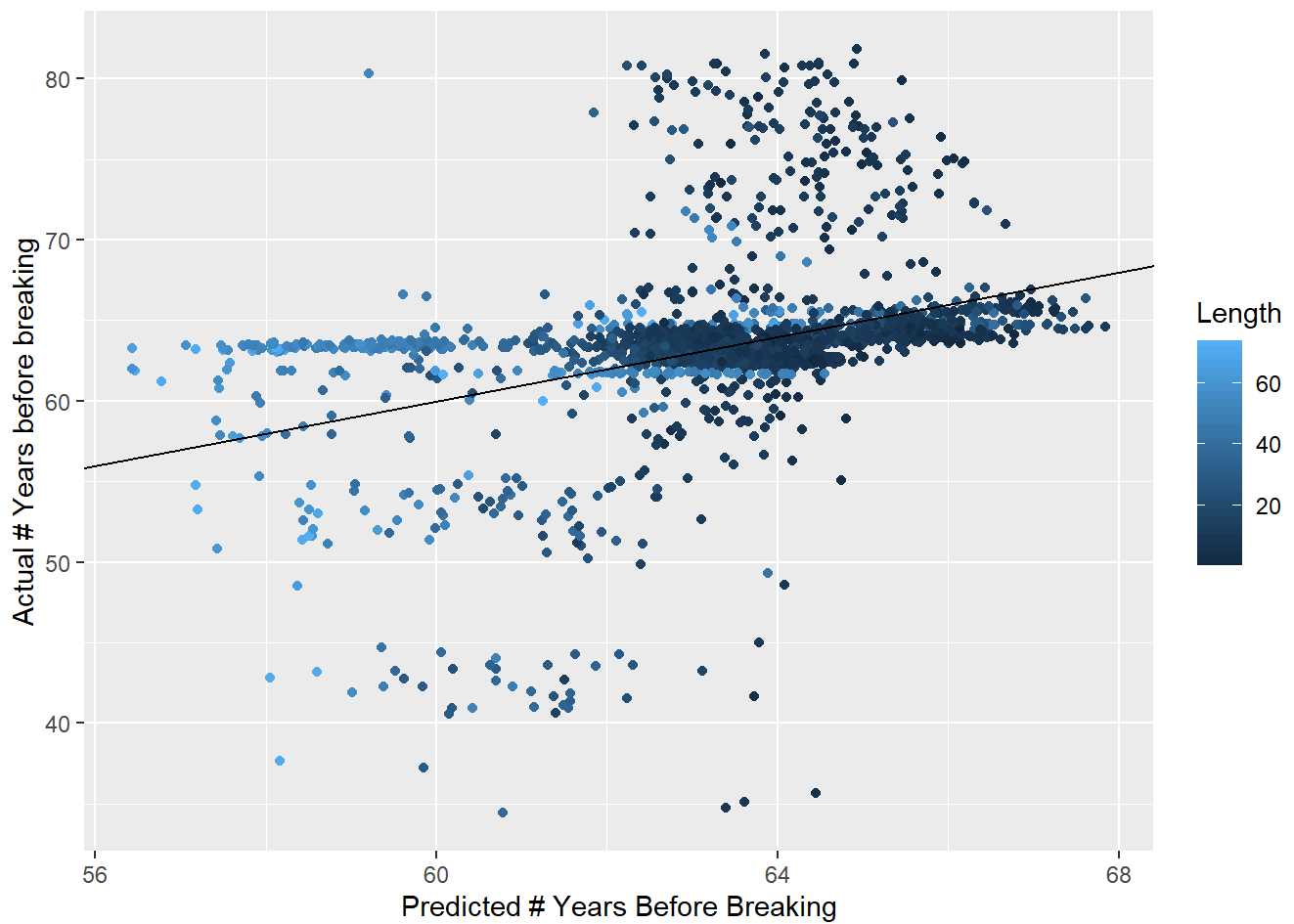
```
ggplot(data=val) +geom_point(mapping=aes(x=pred_v_s1, y=as.numeric(Years),
  colour=Material))+ geom_abline(a=0,b=1)+
labs(x="Predicted # Years Before Breaking", y="Actual # Years before breaking")
```

Warning in geom_abline(a = 0, b = 1): Ignoring unknown parameters: `a` and `b`



```
ggplot(data=val) +geom_point(mapping=aes(x=pred_v_s1, y=as.numeric(Years),  
  colour=length))+ geom_abline(a=0,b=1)+labs(x=  
  "Predicted # Years Before Breaking", y="Actual # Years before breaking")
```

Warning in geom_abline(a = 0, b = 1): Ignoring unknown parameters: `a` and `b`



```
ggplot(data=val) +geom_point(mapping=aes(x=pred_v_s1, y=as.numeric(Years),  
  colour=Surface))+ geom_abline(a=0,b=1)+  
labs(x="Predicted # Years Before Breaking", y="Actual # Years before breaking")
```

Warning in geom_abline(a = 0, b = 1): Ignoring unknown parameters: `a` and `b`

